AKASH KUNDU

AI/ML & Systems Engineer | Computer Vision, Medical Informatics & Edge Runtime

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SUMMARY

Undergraduate Computer Science Engineer specializing in **Computer Vision**, **Biomedical Informatics**, **Model Optimization (ONNX Runtime FP16)**, and **High-Throughput Distributed Systems**. Lead author of **OphthalmoAI** (point-of-care retinal disease screening platform achieving 85.18% accuracy, 0.9818 Macro AUROC, 2.15x serving acceleration) with a peer-reviewed research manuscript in preparation. Experienced in building sovereign/local AI pipelines (Whisper Bengali speech processing, Ollama local inference cascade) and secure microservices. Strong academic foundation with **SGPA 7.93** in the latest completed semester and **95.4% in Class X (CBSE)**.

ACADEMIC RECORD (CLASS X ONWARDS)

Degree / Examination	Institution / Board	Tenure	Score / Grades
B.Tech in CSE (2027 Batch)	Techno India University, Kolkata, West Bengal	Sep 2023 – Present	**SGPA: 7.93** (5th Sem) | **8.29** (4th Sem) CGPA: **7.61 / 10.0** (76.1%)
Class XII (Science)	Arambagh Vivekananda Academy, West Bengal (CBSE)	2021 – 2023	**71.6%** (Physics, Chem, Math, CS)
Class X (Secondary)	Arambagh Vivekananda Academy, West Bengal (CBSE)	2020 – 2021	**95.4%** (Science & Mathematics)

RESEARCH & PUBLICATIONS (IN PREPARATION)

OphthalmoAI: A Calibrated Tri-Backbone Deep Vision Ensemble with Biophysical Guardrails — Manuscript In Preparation

Lead Author: Akash Kundu | Peer-Reviewed Biomedical Informatics Journal Manuscript

- Engineered novel mathematical formulations: **TC-MBE** (Temperature-Calibrated Ensemble), **OAC-DG** (Biophysical Aperture Guardrail), **US-CRC** (Urgency-Stratified Conformal Risk Control), and **PASG-GradCAM** (Pixel-Aligned Saliency Grounding).
- Conducted rigorous multi-cohort evaluation across $n=938$ internal test split and independent external clinical cohorts (IDRiD, India: **91.30% DR recall, 100% PDR recall**; RIM-ONE DL, Spain: **100% fail-safe clinical escalation** on out-of-distribution optical crops).

EXPERIENCE

Purposive AI — Software Engineering Intern (Backend & Systems) — July 2026 – Present | Kolkata, India

- Architected an asynchronous ledger pipeline using FastAPI, Celery, and Whisper, translating conversational **Bengali speech audio** into bank-submittable structured records, cutting manual bookkeeping by **80%**.
- Engineered an automated Sarvam AI to local **Ollama fallback circuit breaker**, eliminating cloud vendor lock-in, slashing inference costs by **87%**, and ensuring sovereign data governance.
- Designed high-throughput task worker queues with PostgreSQL **pgvector** semantic state tracking, sustaining **99.9% uptime** under sub-500ms API turnaround; enforced 100% TDD pass rate across 156 Pytest suites.

KEY TECHNICAL PROJECTS

OphthalmoAI — Point-of-Care Retinal Disease Screening Platform | PyTorch, ONNX, FastAPI, React 19 — GitHub | Archify Map

- Architected a calibrated soft-voting ensemble (**DenseNet-201, ConvNeXt-Small, EfficientNet-V2-M**) with Platt temperature scaling (**0.0381 ECE**), achieving **85.18% accuracy** and **0.9818 Macro AUROC** over 6 retinal disease classes.
- Engineered pre-inference biophysical domain guardrails (**OAC-DG**) evaluating circular aperture geometry and chromophore backscatter ratio ($\rho_{\text{RB}} \ge 1.05$), blocking 100% non-fundus images before GPU allocation.
- Accelerated serving latency to **84.2 ms p50 (2.15x speedup)** at 17.3 QPS via ONNX Runtime FP16 compilation; built offline in-browser edge screening (<50ms, zero cloud egress), PostgreSQL Row-Level Security, and **253 passing automated tests (100% pass rate)**.

C.O.P.P.E.R. — Local AI Operating System & Microservices Engine | FastAPI, Electron, React 19, Python, Docker — GitHub

- Engineered a 30-agent orchestration hierarchy with a Topological Failure-Predicting cascade router executing dispatches at **~9,856 QPS** (<0.10 ms cache dispatch) verified across **538 automated tests**.
- Constructed Forge Sandbox with AST security analysis and 4-tier friction protocol, achieving **100% threat catch sensitivity across 350 adversarial triggers** and Zero-Trust credential firewall.

TECHNICAL SKILLS

AI/ML & Computer Vision:	PyTorch, OpenCV, Deep Vision (DenseNet, ConvNeXt, EfficientNet), Grad-CAM (XAI), Medical Image Processing, Platt Calibration (ECE), Conformal Prediction, scikit-learn
Edge AI & High-Perf Systems:	ONNX Runtime (FP16 Graph Optimization), TensorRT, Distributed Model Serving, Redis, Celery Worker Queues, Asyncio Concurrency
Speech & Vernacular NLP:	Speech-to-Text (Whisper), Bengali Audio Transcription, LLM Orchestration (Ollama, LangChain, LangGraph), Vector Databases (pgvector, ChromaDB)
Backend, Systems & DevOps:	Python, Java, C++, JavaScript, TypeScript, SQL (PostgreSQL RLS, SQLite WAL), FastAPI, React 19, Docker, Linux/Unix CLI, Git, CI/CD Actions, Pytest, Vitest

CERTIFICATIONS

- **Microsoft:**Azure AI Developer Associate (AI-103)
- **Microsoft:**Machine Learning Operations Engineer (AI-300)
- **Oracle:**AI Database Foundations (1Z0-1195)
- **Oracle:**Agentic AI Foundations (1Z0-1157)
- **AICTE:**AI-ML Virtual Internship Accreditation